$$r = \frac{168}{\sqrt{396} \times \sqrt{320}}$$

$$r = \frac{168}{355.98} = 0.472$$

i.e. $r = +0.472$

Heights of fathers and their respective sons are positively correlated.

Example 9.4

Calculate the correlation coefficient from the following data

$N=9$, $\sum X = 45$, $\sum Y = 108$, $\Sigma X^2 = 285$, $\Sigma Y^2 = 1356$, $\Sigma XY = 597$.

Solution:

We know that correlation coefficient

$$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N\Sigma X^2 - (\Sigma X)^2}\sqrt{N\Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{9(597) - (45 \times 108)}{\sqrt{9(285) - (45)^2} \times \sqrt{9(1356) - (108^2)}}$$

$r = +0.95$

Example 9.5

From the following data calculate the correlation coefficient $\Sigma xy = 120$, $\Sigma x^2 = 90$, $\Sigma y^2 = 640$.

Solution:

Given $\Sigma xy = 120$, $\Sigma x^2 = 90$, $\Sigma y^2 = 640$

Then $r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}} = \frac{120}{\sqrt{90(640)}} = \frac{120}{\sqrt{57600}}$

$= \frac{120}{240} = 0.5$

9.2 Rank Correlation

9.2.1 Spearman's Rank Correlation Coefficient

In 1904, Charles Edward Spearman, a British psychologist found out the method of ascertaining the coefficient of correlation by ranks. This method is based on rank. This measure is useful in dealing with qualitative characteristics, such as intelligence, beauty, morality, character, etc. It cannot be measured quantitatively, as in the case of Pearson's coefficient of correlation.

Rank correlation is applicable only to individual observations. The result we get from this method is only an approximate one, because under ranking method original value are not taken into account. The formula for Spearman's rank correlation which is denoted by ρ (pronounced as row) is

$$\rho = 1 - \frac{6\Sigma d^2}{N(N^2 - 1)} \quad \text{(or)}$$

$$\rho = 1 - \frac{6\Sigma d^2}{N^3 - N}$$

where d = The difference of two ranks

$= R_X - R_Y$ and

N = Number of paired observations.

Rank coefficient of correlation value lies between –1 and +1.

Symbolically, $-1 \le \rho \le +1$.

When we come across spearman's rank correlation, we may find two types of problem

(i) When ranks are given

(ii) When ranks are not given

Example 9.6

The following are the ranks obtained by 10 students in Statistics and Mathematics

Statistics	1	2	3	4	5	6	7	8	9	10
Mathematics	1	4	2	5	3	9	7	10	6	8

Find the rank correlation coefficient.

Solution:

Let R_X is considered for the ranks of Statistics and R_Y is considered for the ranks of mathematics.

09_11th_BM-STAT_Ch-9-EM.indd 215 21-04-2020 12:28:19 PM